

SCENARIO 1 - SETTING UP A SMART CAMPUS NETWORK

End-User Layer

Hardware: Laptops, IoT, Phones
Software: OS, client apps.

Access Layer

Hardware: Wireless, APs, L2 Switches
Connection: Multipoint (Wi-Fi)
Network Type: WLAN/LAN

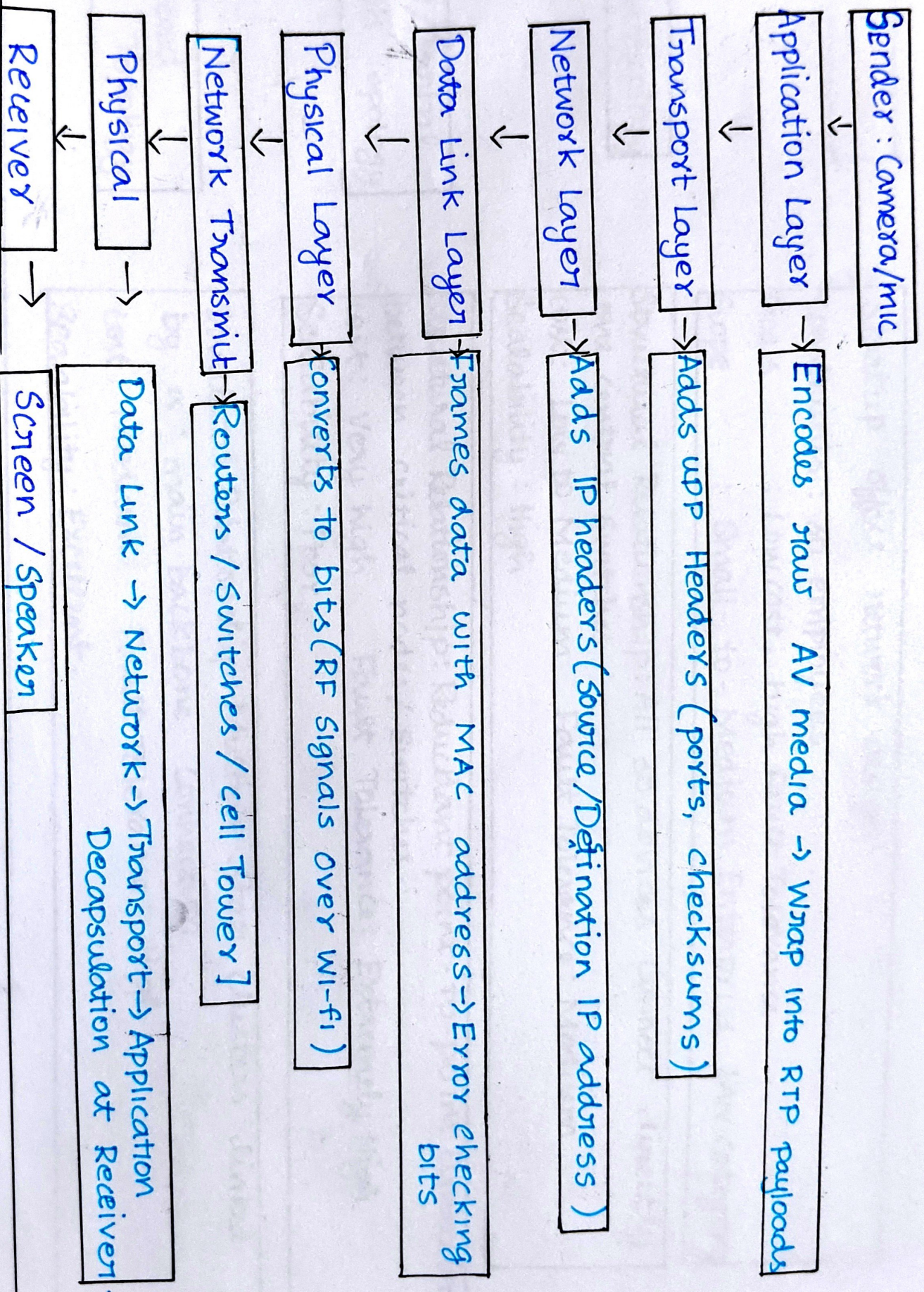
Distribution or
Core Layer

Hardware: L3 Switches, Firewalls
Connection: Point-to-Point (Fiber)
Network type: Campus LAN

Internet Gateway
(Boundary)

Hardware: Core edge Router
Connection: Point-to-Point (Leased)
Network type: WAN Link

SCENARIO 2 - DIAGONAL VIDEO CALL FAILURE



SCENARIO 3 - CHOOSING A TOPOLOGY FOR A NEW START OFFICE

Core Nodes

Startup office network design

constraints : 30 employees

Goals : Low cost, high Fault Tolerance

Scope : Small - to - Medium Enterprise LAN category

Star topology

Structural Relationship: All 30 devices connect directly one central switches

cost: Low to Medium Fault Tolerance: Medium

Scalability : high

Full / Partial

Mesh Topology

Structural Relationship: Redundant point-to-point connections between critical nodes / switches.

cost: Very high Fault Tolerance: Extremely High

Scalability : Poor

Hybrid

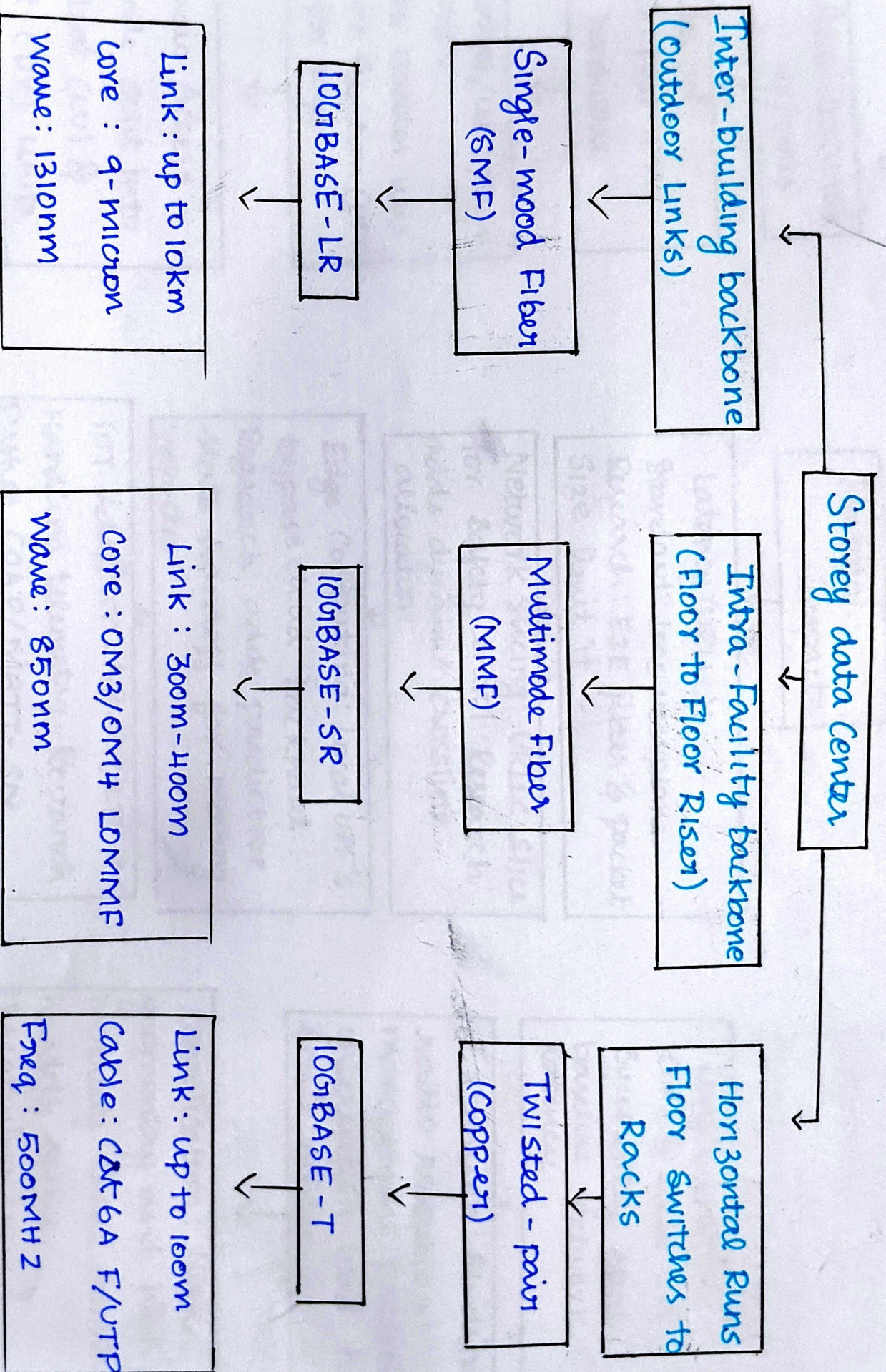
Topology

Structural Relationship: Multiple star clusters linked by a main backbone connection

cost: Medium Fault Tolerance: High

Scalability : Excellent.

SCENARIO 4 - WIRING A MULTI-STOREY DATA CENTER



SCENARIO 5 - ROLLING OUT 5G FOR A SMART TRAFFIC SYSTEM

Architectural
depths

core net:
4G: Flat IPW/
hardware

control/user plane
split:
CUPS isolates user
plane function (UPF)
for edge

Radio Access:
gNode split into
central CU &
Dist. (DU) units

Protocol
Impacts

Latency (URLLC):
Standard: 1ms user plane
Research: E2E jitter & packet
size limit it

Network Slicing: URLLC slice
for safety control. Research
adds dynamic crossline
allocation

Edge Computing: Local UPF's
bypass cloud backhaul.
Research adds predictive
state handoffs for moving
vehicles.

IoT Adaptation: NB-IoT/LTE-M
Handling telemetry. Research
pushes CoAP/MQTT-SN
to reduce headers.

Literature Vs
Standards

Confirmation:
CUPS & local UPF
successfully slash
baseline network
latency

Extension: AI-driven
radio resource
management predicts
intersection load to
scale slides.

Complication: Urban
geometry and high
mobility cause massive
handoff spikes,
threatening URLLC
reliability.